

1. Questions

Study the following data carefully and answer the questions.

There are two solid cubes, A and B, and two solid cuboids, P and Q, and the volume of each of them is different.

The side of cube A is 3 cm more than the side of cube B, and the total surface area of cube B is 64% of that of cube A. The side of cube A is 60% of the length of cuboid P; the breadth of cuboid P is 60% of the side of cube A, and the numerical value of the volume (in cm^3) of cuboid P is equal to the numerical value of the total surface area (in cm^2) of cube A. The side of cube B is 60% of the length of cuboid Q; the breadth of cuboid Q is $66\frac{2}{3}\%$ of the side of cube B, and the numerical value of the volume (in cm^3) of cuboid Q is 96 more than the numerical value of the total surface area (in cm^2) of cube B.

Cube A is melted down, and M number of small solid cubes (each of side 5 cm) are formed. Cuboid Q is melted down, and N number of small solid cubes (each of side 4 cm) are formed. Find the ratio of M to N?

- a. 3: 2
- b. 54: 25
- c. 9: 5
- d. 18: 13
- e. None of these

2. Questions

If the cost of painting is Rs($5M - 4N$) per cm^2 , then find the total cost of painting cube B and cuboid P together. (Take the values of M and N from the above question.)

- a. Rs.121950
- b. Rs.129150
- c. Rs.121590
- d. Rs.125190
- e. Rs.129510

@QuizByMickey

3. Questions

If the ratio of the volume of cuboid P to the volume of cuboid Q is R: S, then find which of the following statements is/are true.

X: The difference of R and S is a prime number.

Y: The value of $(8R - 2S)$ is a perfect square.

Z: The ratio of the length of cuboid P to that of cuboid Q is $(R/9): (S/8)$.

- a. Only X and Z
- b. All are true
- c. Only X
- d. Only Y and Z
- e. Only Y

4. Questions

If cuboids P and Q are melted down and a solid cuboid is formed, whose length is $(66 \frac{2}{3})\%$ of the sum of the lengths of cuboids P and Q and whose breadth is 6 cm less than the sum of the breadths of cuboids P and Q, then find that the height of the new cuboid is what percentage more/less than that of cuboid P?

- a. $(13 \frac{1}{3})\%$
- b. 15%
- c. $(11 \frac{1}{9})\%$
- d. $(9 \frac{1}{11})\%$
- e. $(16 \frac{2}{3})\%$

5. Questions

If the side of cube C is 60% of that of cube A and the length, breadth, and height of cuboid R were respectively 45%, 75%, and 50% of the length, breadth, and height of cuboid Q, then what would have been the ratio of the volume of cube C to that of cuboid R?

- a. 27: 14
- b. 21: 13
- c. 36: 25
- d. 9: 2
- e. 3: 1

6. Questions

@QuizByMickey

Study the following data carefully and answer the questions.

The data given below is related to the amounts invested by a person in five schemes (A, B, C, D, and E) in two years, and the rate of simple interest in each scheme for each year is different.

Bar graph given below shows the following data:

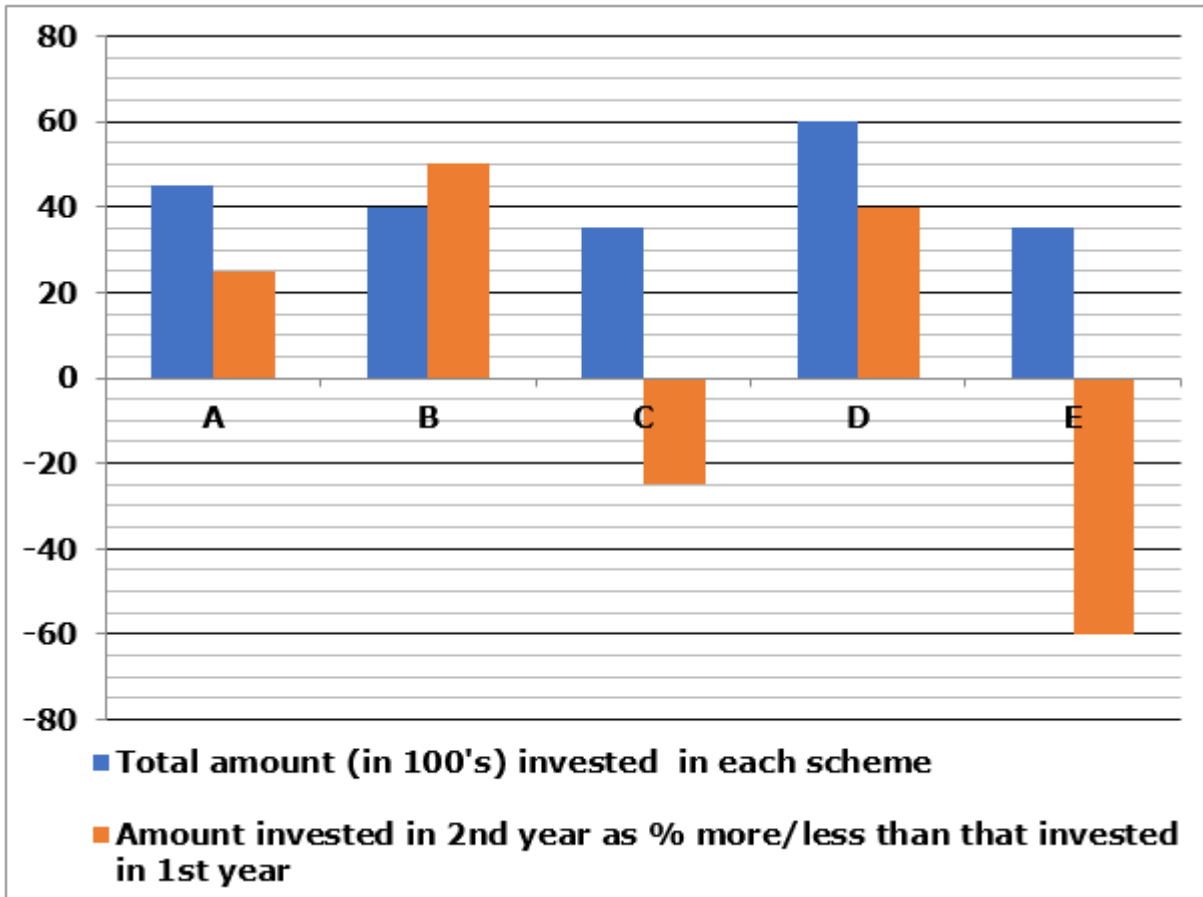


Table given below shows the following data:

Scheme	The interest received after 1 year (in ₹)	(Rate of interest for 1 st year) – (Rate of interest for 2 nd year) (in %)
A	170	-3.5
B	240	10
C	150	-2.5
D	150	-6
E	200	-6

Note:

- A positive number indicates a percentage more, and a negative number indicates a percentage less.
- The total amount in the scheme in the second year = An amount invested in the 1st year + An amount invested in the 2nd year

If the rates of interest in scheme B for both years were interchanged, then the total simple interest received from scheme B after 2 years would have been what approximate percentage more/less than the original simple interest received from scheme B after 2 years?

- 45.5%
- 65.5%

@QuizByMickey

- c. 54.5%
- d. 48.5%
- e. 60.5%

7. Questions

If the total simple interest received from scheme A after 2 years is Rs.M, then the total simple interest received from scheme D after 2 years will lie between which of the following ranges?

P: Between $(M + 100)$ and $(M + 200)$

Q: Between $(M + 160)$ and $(M + 240)$

R: Between $(M + 40)$ and $(M + 150)$

- a. Only P
- b. Only P and R
- c. Only Q and R
- d. Only R
- e. Only Q

8. Questions

If the total amount invested in all five schemes together in the 1st year were invested in scheme F in the 1st year and the total amount invested in all five schemes together in the 2nd year were invested in scheme F in the 2nd year, then what would have been the total simple interest received from scheme F after 2 years?

Note: The interest rate for the 1st year and 2nd year is the same as scheme E.

- a. Rs.3355
- b. Rs.4286
- c. Rs.3858
- d. Rs.4045
- e. Rs.4488

@QuizByMickey

9. Questions

If the interest rates of all schemes for the 1st year are described in a circle and the difference between the degree distributions of the interest rates of schemes A and B for the 1st year is N° , then find the total degree distribution of the interest rates of schemes C, D, and E together for the 1st year.

- a. $(4N - 48)^\circ$
- b. $(3N + 16)^\circ$

- c. $(2N + 76)^\circ$
- d. None of these
- e. $(5N - 78)^\circ$

10. Questions

From the above question, find that which of the following statements is/are true?

X: The simple interest received from scheme B in the 2nd year is Rs.[$4(N - 2)$].

Y: The difference between the simple interest received from scheme C after 1 year and that received from scheme C after 2 years is Rs.($3N + 44$).

Z: The sum of the interest rates of all five schemes for the 2nd year is $(N + 1)\%$.

- a. None is true
- b. Only X and Z
- c. All are true
- d. Only Y and Z
- e. Only X and Y

11. Questions

Bikes A, B, and C are running in the same direction such that bike B is following bike C, bike C is following bike A, and the speed of bike A is 64.8 km/h. The distance between the bikes A and C is 160 m, and the distance between the bikes B and C is 120. If bike B crosses bike A in 40 seconds, then find that which of the following can't be determined?

P: The time taken by bike B to cross a 450 m long bridge.

Q: The time taken by bike B to cross bike C.

R: The time taken by bike C to cross bike A.

- a. Only Q
- b. Only P and R
- c. Only R
- d. Only Q and R
- e. Only P

@QuizByMickey

12. Questions

A shopkeeper marked up an article by 60% and received Rs.25 profit by selling at __% discount on its marked price. If he had sold the article at 40% profit and sold the article at 20% loss, then his total profit would have been Rs.25. Find the value that will be filled in the blank.

- a. Can't be determined

- b. 12.5
- c. 40
- d. 30
- e. 25

13. Questions

In a mixture of milk and water, the quantity of milk is 40 L more than that of water, and the cost of pure milk is Rs.72 per L. A profit of 25% is received when the mixture is sold at Rs.81 per L. When ___ L water is added to the mixture, the cost of the mixture becomes Rs.___ per L.

Find which of the following can't be filled in the blank in the same order.

P: 10, 54

Q: 4, 60

R: 14, 48

- a. Only R
- b. Only Q and R
- c. Only P and R
- d. Only Q
- e. Only P and Q

14. Questions

There are three members in a family: A, B, and C, and the average age of the family is 24 years. The difference between A's and B's present ages is equal to C's present age, and A is elder to B.

I: If B's present age is 30 years, then C's age after 6 years will be 'P' years.

II: If B's present age is 20 years, then C's age 8 years ago was 'Q' years.

III: If B's present age is 24 years, then C's present age will be 'R' years.

Find the ratio between P, Q, and R, respectively?

- a. 6: 4: 5
- b. 3: 2: 3
- c. 2: 1: 2
- d. 3: 2: 4
- e. 2: 1: 1

@QuizByMickey

15. Questions

A alone can complete a work in 18 hours, and B alone can complete the work in 45 hours. A and B

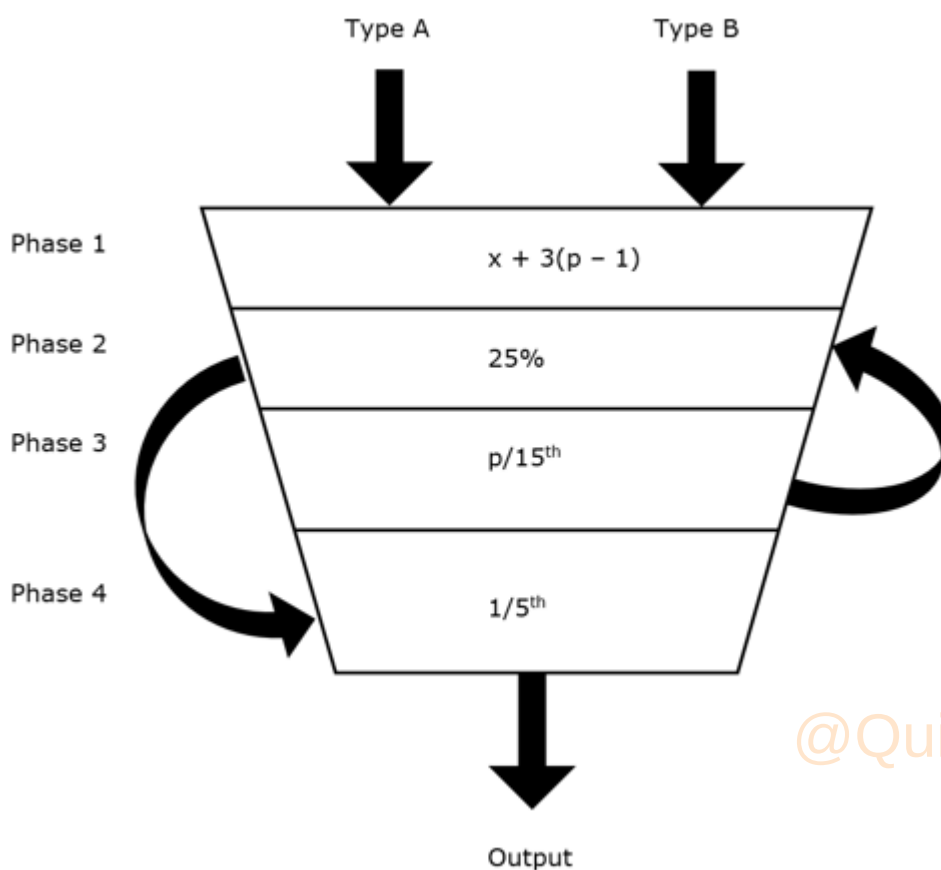
started the work together. After 't' hours, A left the work, and B alone completed the remaining work in 10 hours. If C alone can complete the work in '3t' hours, then in what time can A, B, and C together complete the work?

- a. $(t/2)$ hours
- b. $(t - 1)$ hours
- c. $(t + 5)$ hours
- d. $(2t)$ hours
- e. $(t - 4)$ hours

16. Questions

Study the following data carefully and answer the questions.

All the manufactured units of two types, A and B, by a company underwent four testing phases simultaneously. Type A manufactured is '23p + 3' more than type B, while the ratio of the manufactured number of type A to type B is 3:2. Testing phases are given below in a funnel chart:



@QuizByMickey

Note:

- 1). Order of the phase to be followed: Phase 1 -> Phase 2 (Round 1) -> Phase 3 -> Phase 2 (Round 2) -> Phase 4.
- 2). Data mentioned in each phase is the total number of units of both types of items that are discarded in that phase.

3). Number of units reached in any phase of the testing = Number of units reached for testing in the previous phase – Number of units discarded in the previous phase.

4). The total number of output units is 132.

5). 'x' (< 100) is an even multiple of 11 and two more than a perfect square of an integer.

What is the total number of discarded units of both types of items together from phase 1 to phase 2 (Round 2) of the testing?

- a. 325
- b. 343
- c. 310
- d. 268
- e. None of these

17. Questions

If 'x' % of total units passed in phase 2 (Round 1) of the testing are of type A, then find the difference between total units of type A and type B that are passed in that phase.

- a. 90
- b. 108
- c. 120
- d. 84
- e. 96

18. Questions

If there is one more phase, phase 5, after phase 4 of the testing in which total 'y' units are discarded ('y' is a 2-digit (> 20) prime number '1' more than a perfect square), then what percent of total manufactured items by the company passed all five phases of testing?

- a. 25%
- b. 20%
- c. 30%
- d. $33\frac{1}{3}\%$
- e. $16\frac{2}{3}\%$

@QuizByMickey

19. Questions

If in phase 1, the ratio of passed units of item A to item B is 11: 9, then find the ratio of discarded units of item A to item B in phase 1.

- a. 8: 3

- b. 9: 4
- c. 15: 4
- d. 11: 2
- e. 13: 2

20. Questions

If total passed units of both the items in phase 4 of the testing is $(10a + b)/100$ part of total passed units of both the items in phase 1, then find the simplest value of $(a + b)/ab$.

Note: Both a and b are the single digit number.

- a. $2/3$
- b. $4/5$
- c. $3/5$
- d. $1/3$
- e. $5/8$

Explanations:

1. Questions

Let the side of cube B = 'x' cm

So, the side of cube A = $(x + 3)$ cm

Since, the total surface area of cube B is 64% of that of cube A.

So,

$$6x^2 = 6(x + 3)^2 * (64/100)$$

$$x^2 = (x + 3)^2 * (16/25)$$

$$25x^2 = 16(x^2 + 9 + 6x)$$

$$25x^2 = 16x^2 + 144 + 96x$$

$$9x^2 - 96x - 144 = 0$$

$$3x^2 - 32x - 48 = 0$$

$$3x^2 - 36x + 4x - 48 = 0$$

$$3x(x - 12) + 4(x - 12) = 0$$

$$x = 12$$

@QuizByMickey

The side of cube A = $12 + 3 = 15$ cm

The side of cube B = 12 cm

Since, the side of cube A is 60% of the length of cuboid P.

So, the length of cuboid P = $15 * (100/60) = 25$ cm

Since, the breadth of cuboid P is 60% of the side of cube A.

So, the breadth of cuboid P = 60% of 15 = 9 cm

Since, the numerical value of the volume (in cm^3) of cuboid P is equal to the numerical value of the total surface area (in cm^2) of cube A.

And the total surface area of cube A = $6 * 15^2 = 1350 \text{ cm}^2$

So, the volume of cuboid P = 1350 cm^3

And the height of cuboid P = $1350/(25 * 9) = 6$ cm

Since, the side of cube B is 60% of the length of cuboid Q.

So, the length of cuboid Q = $12 * (100/60) = 20$ cm

Since, the breadth of cuboid Q is $66\frac{2}{3}\%$ of the side of cube B.

So, the breadth of cuboid Q = $66\frac{2}{3}\%$ of 12 = 8 cm

Since, the numerical value of the volume (in cm^3) of cuboid Q is 96 more than the numerical value of the total surface area (in cm^2) of cube B.

And the total surface area of cube B = $6 * 12^2 = 864 \text{ cm}^2$

So, the volume of cuboid Q = $864 + 96 = 960 \text{ cm}^3$

And the height of cuboid Q = $960/(20 * 8) = 6$ cm

The side of cube A = 15 cm

The side of cube B = 12 cm

The length of cuboid P = 25 cm

The breadth of cuboid P = 9 cm

The height of cuboid P = 6 cm

The length of cuboid Q = 20 cm

The breadth of cuboid Q = 8 cm

The height of cuboid Q = 6 cm

Answer: C

Since, the side of each cube formed by melting cube A = 5 cm

@QuizByMickey

And the volume of cube A = $15 * 15 * 15 = 3375 \text{ cm}^3$

So, the value of M = $3375/5^3 = 27$

Since, the side of each cube formed by melting cuboid Q = 4 cm

And the volume of cuboid Q = $20 * 8 * 6 = 960 \text{ cm}^3$

So, the value of N = $960/4^3 = 15$

Required ratio = M: N = 27: 15 = 9: 5

2. Questions

Let the side of cube B = 'x' cm

So, the side of cube A = (x + 3) cm

Since, the total surface area of cube B is 64% of that of cube A.

So,

$$6x^2 = 6(x + 3)^2 * (64/100)$$

$$x^2 = (x + 3)^2 * (16/25)$$

$$25x^2 = 16(x^2 + 9 + 6x)$$

$$25x^2 = 16x^2 + 144 + 96x$$

$$9x^2 - 96x - 144 = 0$$

$$3x^2 - 32x - 48 = 0$$

$$3x^2 - 36x + 4x - 48 = 0$$

$$3x(x - 12) + 4(x - 12) = 0$$

$$x = 12$$

The side of cube A = $12 + 3 = 15 \text{ cm}$

The side of cube B = 12 cm

Since, the side of cube A is 60% of the length of cuboid P.

So, the length of cuboid P = $15 * (100/60) = 25 \text{ cm}$

Since, the breadth of cuboid P is 60% of the side of cube A.

So, the breadth of cuboid P = 60% of 15 = 9 cm

Since, the numerical value of the volume (in cm^3) of cuboid P is equal to the numerical value of the total surface area (in cm^2) of cube A.

@QuizByMickey

And the total surface area of cube A = $6 * 15^2 = 1350 \text{ cm}^2$

So, the volume of cuboid P = 1350 cm^3

And the height of cuboid P = $1350 / (25 * 9) = 6 \text{ cm}$

Since, the side of cube B is 60% of the length of cuboid Q.

So, the length of cuboid Q = $12 * (100/60) = 20 \text{ cm}$

Since, the breadth of cuboid Q is $66\frac{2}{3}\%$ of the side of cube B.

So, the breadth of cuboid Q = $66\frac{2}{3}\%$ of 12 = 8 cm

Since, the numerical value of the volume (in cm^3) of cuboid Q is 96 more than the numerical value of the total surface area (in cm^2) of cube B.

And the total surface area of cube B = $6 * 12^2 = 864 \text{ cm}^2$

So, the volume of cuboid Q = $864 + 96 = 960 \text{ cm}^3$

And the height of cuboid Q = $960 / (20 * 8) = 6 \text{ cm}$

The side of cube A = 15 cm

The side of cube B = 12 cm

The length of cuboid P = 25 cm

The breadth of cuboid P = 9 cm

The height of cuboid P = 6 cm

The length of cuboid Q = 20 cm

The breadth of cuboid Q = 8 cm

The height of cuboid Q = 6 cm

Answer: B

From the above question:

Since, the value of M = 27

And the value of N = 15

So, the cost of painting = $(5 * 27 - 4 * 15) = \text{Rs.}75 \text{ cm}^2$

Since, the total surface area of cube B = $6 * 12^2 = 864 \text{ cm}^2$

And the total surface area of cuboid P = $2(25 * 9 + 9 * 6 + 25 * 6) = 858 \text{ cm}^2$

So, the required total cost of painting = $(864 + 858) * 75 = \text{Rs.}129150$

3. Questions

@QuizByMickey

Let the side of cube B = 'x' cm

So, the side of cube A = (x + 3) cm

Since, the total surface area of cube B is 64% of that of cube A.

So,

$$6x^2 = 6(x + 3)^2 * (64/100)$$

$$x^2 = (x + 3)^2 * (16/25)$$

$$25x^2 = 16(x^2 + 9 + 6x)$$

$$25x^2 = 16x^2 + 144 + 96x$$

$$9x^2 - 96x - 144 = 0$$

$$3x^2 - 32x - 48 = 0$$

$$3x^2 - 36x + 4x - 48 = 0$$

$$3x(x - 12) + 4(x - 12) = 0$$

$$x = 12$$

The side of cube A = 12 + 3 = 15 cm

The side of cube B = 12 cm

Since, the side of cube A is 60% of the length of cuboid P.

So, the length of cuboid P = 15 * (100/60) = 25 cm

Since, the breadth of cuboid P is 60% of the side of cube A.

So, the breadth of cuboid P = 60% of 15 = 9 cm

Since, the numerical value of the volume (in cm³) of cuboid P is equal to the numerical value of the total surface area (in cm²) of cube A.

And the total surface area of cube A = 6 * 15² = 1350 cm²

So, the volume of cuboid P = 1350 cm³

And the height of cuboid P = 1350/(25 * 9) = 6 cm

Since, the side of cube B is 60% of the length of cuboid Q.

So, the length of cuboid Q = 12 * (100/60) = 20 cm

Since, the breadth of cuboid Q is 66(2/3)% of the side of cube B.

So, the breadth of cuboid Q = 66(2/3)% of 12 = 8 cm

@QuizByMickey

Since, the numerical value of the volume (in cm^3) of cuboid Q is 96 more than the numerical value of the total surface area (in cm^2) of cube B.

And the total surface area of cube B = $6 * 12^2 = 864 \text{ cm}^2$

So, the volume of cuboid Q = $864 + 96 = 960 \text{ cm}^3$

And the height of cuboid Q = $960 / (20 * 8) = 6 \text{ cm}$

The side of cube A = 15 cm

The side of cube B = 12 cm

The length of cuboid P = 25 cm

The breadth of cuboid P = 9 cm

The height of cuboid P = 6 cm

The length of cuboid Q = 20 cm

The breadth of cuboid Q = 8 cm

The height of cuboid Q = 6 cm

Answer: A

Since, the volume of cuboid P = $25 * 9 * 6 = 1350 \text{ cm}^3$

And the volume of cuboid Q = $20 * 8 * 6 = 960 \text{ cm}^3$

So, the value of R: S = $1350: 960 = 45: 32$

From X:

The difference between R and S = $45 - 32 = 13$

Since, 13 is a prime number.

So, X is true.

From Y:

The value of $(8R - 2S) = (8 * 45 - 2 * 32) = 296$

Since, 296 is not a perfect square.

So, Y is not true.

From Z:

The ratio of the length of cuboid P to that of cuboid Q = $25: 20 = 5: 4$

Since, the value of $(R/9): (S/8) = (45/9): (32/8) = 5: 4$

So, Z is true.

Hence, only X and Z are true.

@QuizByMickey

4. Questions

Let the side of cube B = 'x' cm

So, the side of cube A = (x + 3) cm

Since, the total surface area of cube B is 64% of that of cube A.

So,

$$6x^2 = 6(x + 3)^2 * (64/100)$$

$$x^2 = (x + 3)^2 * (16/25)$$

$$25x^2 = 16(x^2 + 9 + 6x)$$

$$25x^2 = 16x^2 + 144 + 96x$$

$$9x^2 - 96x - 144 = 0$$

$$3x^2 - 32x - 48 = 0$$

$$3x^2 - 36x + 4x - 48 = 0$$

$$3x(x - 12) + 4(x - 12) = 0$$

$$x = 12$$

The side of cube A = 12 + 3 = 15 cm

The side of cube B = 12 cm

Since, the side of cube A is 60% of the length of cuboid P.

So, the length of cuboid P = 15 * (100/60) = 25 cm

Since, the breadth of cuboid P is 60% of the side of cube A.

So, the breadth of cuboid P = 60% of 15 = 9 cm

Since, the numerical value of the volume (in cm³) of cuboid P is equal to the numerical value of the total surface area (in cm²) of cube A.

And the total surface area of cube A = 6 * 15² = 1350 cm²

So, the volume of cuboid P = 1350 cm³

And the height of cuboid P = 1350/(25 * 9) = 6 cm

Since, the side of cube B is 60% of the length of cuboid Q.

So, the length of cuboid Q = 12 * (100/60) = 20 cm

Since, the breadth of cuboid Q is 66(2/3)% of the side of cube B.

So, the breadth of cuboid Q = 66(2/3)% of 12 = 8 cm

@QuizByMickey

Since, the numerical value of the volume (in cm^3) of cuboid Q is 96 more than the numerical value of the total surface area (in cm^2) of cube B.

And the total surface area of cube B = $6 * 12^2 = 864 \text{ cm}^2$

So, the volume of cuboid Q = $864 + 96 = 960 \text{ cm}^3$

And the height of cuboid Q = $960 / (20 * 8) = 6 \text{ cm}$

The side of cube A = 15 cm

The side of cube B = 12 cm

The length of cuboid P = 25 cm

The breadth of cuboid P = 9 cm

The height of cuboid P = 6 cm

The length of cuboid Q = 20 cm

The breadth of cuboid Q = 8 cm

The height of cuboid Q = 6 cm

Answer: E

Since, the total length of cuboids P and Q together = $25 + 20 = 45 \text{ cm}$

So, the length of new cuboid = $66\frac{2}{3}\%$ of $45 = 30 \text{ cm}$

Since, the total breadth of cuboid P and Q together = $9 + 8 = 17 \text{ cm}$

So, the breadth of new cuboid = $17 - 6 = 11 \text{ cm}$

Since, the total volume of cuboids P and Q together:

$$= (25 * 9 * 6) + (20 * 8 * 6)$$

$$= 1350 + 960$$

$$= 2310 \text{ cm}^3$$

So, the volume of new cuboid = 2310 cm^3

And the height of new cuboid = $2310 / (30 * 11) = 7 \text{ cm}$

Since, the height of cuboid P = 6 cm

So, the required percentage = $[(7 - 6) / 6] * 100 = (16 \frac{2}{3})\%$

5. Questions

Let the side of cube B = 'x' cm

So, the side of cube A = $(x + 3) \text{ cm}$

Since, the total surface area of cube B is 64% of that of cube A.

@QuizByMickey

So,

$$6x^2 = 6(x + 3)^2 * (64/100)$$

$$x^2 = (x + 3)^2 * (16/25)$$

$$25x^2 = 16(x^2 + 9 + 6x)$$

$$25x^2 = 16x^2 + 144 + 96x$$

$$9x^2 - 96x - 144 = 0$$

$$3x^2 - 32x - 48 = 0$$

$$3x^2 - 36x + 4x - 48 = 0$$

$$3x(x - 12) + 4(x - 12) = 0$$

$$x = 12$$

The side of cube A = $12 + 3 = 15$ cm

The side of cube B = 12 cm

Since, the side of cube A is 60% of the length of cuboid P.

So, the length of cuboid P = $15 * (100/60) = 25$ cm

Since, the breadth of cuboid P is 60% of the side of cube A.

So, the breadth of cuboid P = 60% of 15 = 9 cm

Since, the numerical value of the volume (in cm^3) of cuboid P is equal to the numerical value of the total surface area (in cm^2) of cube A.

And the total surface area of cube A = $6 * 15^2 = 1350 \text{ cm}^2$

So, the volume of cuboid P = 1350 cm^3

And the height of cuboid P = $1350 / (25 * 9) = 6$ cm

Since, the side of cube B is 60% of the length of cuboid Q.

So, the length of cuboid Q = $12 * (100/60) = 20$ cm

Since, the breadth of cuboid Q is $66\frac{2}{3}\%$ of the side of cube B.

So, the breadth of cuboid Q = $66\frac{2}{3}\%$ of 12 = 8 cm

Since, the numerical value of the volume (in cm^3) of cuboid Q is 96 more than the numerical value of the total surface area (in cm^2) of cube B.

And the total surface area of cube B = $6 * 12^2 = 864 \text{ cm}^2$

@QuizByMickey

So, the volume of cuboid Q = $864 + 96 = 960 \text{ cm}^3$

And the height of cuboid Q = $960 / (20 * 8) = 6 \text{ cm}$

The side of cube A = 15 cm

The side of cube B = 12 cm

The length of cuboid P = 25 cm

The breadth of cuboid P = 9 cm

The height of cuboid P = 6 cm

The length of cuboid Q = 20 cm

The breadth of cuboid Q = 8 cm

The height of cuboid Q = 6 cm

Answer: D

Since, the side of cube A = 15 cm

So, the side of cube C = 60% of 15 = 9 cm

And the volume of cube C = $9^3 = 729 \text{ cm}^3$

Since, the length of cuboid Q = 20 cm

So, the length of cuboid R will be = 45% of 20 = 9 cm

Since, the breadth of cuboid Q = 8 cm

So, the breadth of cuboid R will be = 75% of 8 = 6 cm

Since, the height of cuboid Q = 6 cm

So, the height of cuboid R will be = 50% of 6 = 3 cm

And the new volume of cuboid Q will be = $9 * 6 * 3 = 162 \text{ cm}^3$

Required ratio = 729: 162 = 9: 2

6. Questions

Scheme A:

@QuizByMickey

The total amount in both the years together = Rs.4500

Since, the amount invested in 2nd year is 25% more than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = 100: 125 = 4: 5

The amount invested in 1st year = $4500 * (4/9) = \text{Rs.}2000$

And the amount invested in 2nd year = $4500 - 2000 = \text{Rs.}2500$

Let the rate of interest for 1st year = a%

So,

$$(2000 * a)/100 = 170$$

$$a = 8.5\%$$

Since, the rate of interest for 1st year = 8.5%

So, the rate of interest for 2nd year = 8.5 + 3.5 = 12%

Scheme B:

The total amount in both the years together = Rs.4000

Since, the amount invested in 2nd year is 50% more than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = 100: 150 = 2: 3

The amount invested in 1st year = 4000 * (5/2) = Rs.1600

And the amount invested in 2nd year = 4000 – 1600 = Rs.2400

Let the rate of interest for 1st year = b%

So,

$$(1600 * b)/100 = 240$$

$$b = 15\%$$

Since, the rate of interest for 1st year = 15%

So, the rate of interest for 2nd year = 15 - 10 = 5%

Scheme C:

The total amount in both the years together = Rs.3500

Since, the amount invested in 2nd year is 25% less than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = 100: 75 = 4: 3

The amount invested in 1st year = 3500 * (4/7) = Rs.2000

And the amount invested in 2nd year = 3500 – 2000 = Rs.1500

Let the rate of interest for 1st year = c%

So,

$$(2000 * c)/100 = 150$$

$$c = 7.5\%$$

Since, the rate of interest for 1st year = 7.5%

So, the rate of interest for 2nd year = 7.5 + 2.5 = 10%

Scheme D:

The total amount in both the years together = Rs.6000

Since, the amount invested in 2nd year is 40% more than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = 100: 140 = 5: 7

The amount invested in 1st year = 6000 * (5/12) = Rs.2500

And the amount invested in 2nd year = 6000 – 2500 = Rs.3500

Let the rate of interest for 1st year = d%

So,

$$(2500 * d)/100 = 150$$

$$d = 6\%$$

Since, the rate of interest for 1st year = 6%

So, the rate of interest for 2nd year = 6 + 6 = 12%

Scheme E:

The total amount in both the years together = Rs.3500

Since, the amount invested in 2nd year is 60% loss than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = 100: 40 = 5: 2

The amount invested in 1st year = 3500 * (5/7) = Rs.2500

And the amount invested in 2nd year = 3500 – 2500 = Rs.1000

Let the rate of interest for 1st year = e%

So,

$$(2500 * e)/100 = 200$$

$$e = 8\%$$

Since, the rate of interest for 1st year = 8%

So, the rate of interest for 2nd year = 8 + 6 = 14%

@QuizByMickey

So, the table

Scheme	An amount invested in 1 st year (in ₹)	An amount invested in 2 nd year (in ₹)	The rate of interest for 1 st year	The rate of interest for 2 nd year
A	2000	2500	8.5%	12%
B	1600	2400	15%	5%
C	2000	1500	7.5%	10%
D	2500	3500	6%	12%
E	2500	1000	8%	14%

Answer: C

The total SI received from scheme B after 2 years:

$$= [(1600 * 15)/100] + [(4000 * 5)/100]$$

$$= 240 + 200$$

$$= \text{Rs.}440$$

When the rates of interest for both years were interchanged:

The new total SI received from scheme B after 2 years:

$$= [(1600 * 5)/100] + [(4000 * 15)/100]$$

$$= 80 + 600$$

$$= \text{Rs.}680$$

$$\text{Required percentage} = [(680 - 440)/440] * 100 = 54.5\% \text{ (approx.)}$$

7. Questions

Scheme A:

The total amount in both the years together = Rs.4500

Since, the amount invested in 2nd year is 25% more than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = 100: 125 = 4: 5

The amount invested in 1st year = $4500 * (4/9) = \text{Rs.}2000$

And the amount invested in 2nd year = $4500 - 2000 = \text{Rs.}2500$

Let the rate of interest for 1st year = a%

So,

$$(2000 * a)/100 = 170$$

$$a = 8.5\%$$

Since, the rate of interest for 1st year = 8.5%

So, the rate of interest for 2nd year = $8.5 + 3.5 = 12\%$

Scheme B:

The total amount in both the years together = Rs.4000

Since, the amount invested in 2nd year is 50% more than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = 100: 150 = 2: 3

The amount invested in 1st year = $4000 * (5/2) = \text{Rs.}1600$

And the amount invested in 2nd year = $4000 - 1600 = \text{Rs.}2400$

Let the rate of interest for 1st year = b%

So,

$$(1600 * b)/100 = 240$$

$$b = 15\%$$

Since, the rate of interest for 1st year = 15%

So, the rate of interest for 2nd year = $15 - 10 = 5\%$

Scheme C:

The total amount in both the years together = Rs.3500

Since, the amount invested in 2nd year is 25% less than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = 100: 75 = 4: 3

The amount invested in 1st year = $3500 * (4/7) = \text{Rs.}2000$

And the amount invested in 2nd year = $3500 - 2000 = \text{Rs.}1500$

Let the rate of interest for 1st year = c%

So,

$$(2000 * c)/100 = 150$$

$$c = 7.5\%$$

Since, the rate of interest for 1st year = 7.5%

So, the rate of interest for 2nd year = $7.5 + 2.5 = 10\%$

@QuizByMickey

Scheme D:

The total amount in both the years together = Rs.6000

Since, the amount invested in 2nd year is 40% more than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = 100: 140 = 5: 7

The amount invested in 1st year = $6000 * (5/12) = \text{Rs.}2500$

And the amount invested in 2nd year = $6000 - 2500 = \text{Rs.}3500$

Let the rate of interest for 1st year = d%

So,

$$(2500 * d)/100 = 150$$

$$d = 6\%$$

Since, the rate of interest for 1st year = 6%

So, the rate of interest for 2nd year = $6 + 6 = 12\%$

Scheme E:

The total amount in both the years together = Rs.3500

Since, the amount invested in 2nd year is 60% loss than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = 100: 40 = 5: 2

The amount invested in 1st year = $3500 * (5/7) = \text{Rs.}2500$

And the amount invested in 2nd year = $3500 - 2500 = \text{Rs.}1000$

Let the rate of interest for 1st year = e%

So,

$$(2500 * e)/100 = 200$$

$$e = 8\%$$

Since, the rate of interest for 1st year = 8%

So, the rate of interest for 2nd year = $8 + 6 = 14\%$

So, the table

@QuizByMickey

Scheme	An amount invested in 1 st year (in ₹)	An amount invested in 2 nd year (in ₹)	The rate of interest for 1 st year	The rate of interest for 2 nd year
A	2000	2500	8.5%	12%
B	1600	2400	15%	5%
C	2000	1500	7.5%	10%
D	2500	3500	6%	12%
E	2500	1000	8%	14%

Answer: A

Since, the total SI received from scheme A after 2 years:

$$= [(2000 * 8.5)/100] + [(2500 * 12)/100]$$

$$= 170 + 300$$

$$= \text{Rs.} 470$$

$$\text{So, } M = 470$$

Now, the total SI received from scheme D after 2 years:

$$= [(2500 * 6)/100] + [(3500 * 12)/100]$$

$$= 150 + 420$$

$$= \text{Rs.} 570$$

From P:

$$\text{The value of } (M + 100) = 470 + 100 = 570$$

$$\text{The value of } (M + 200) = 470 + 200 = 670$$

Since, 570 lies between 570 and 670.

So, P is true.

From Q:

$$\text{The value of } (M + 160) = 470 + 160 = 630$$

$$\text{The value of } (M + 240) = 470 + 240 = 710$$

Since, 630 doesn't lie between 630 and 710.

So, Q is not true.

From R:

$$\text{The value of } (M + 40) = 470 + 40 = 510$$

$$\text{The value of } (M + 150) = 470 + 150 = 620$$

@QuizByMickey

Since, 870 doesn't lie between 750 and 860.

So, R is not true.

Hence, required value lies between the numbers given in P only.

8. Questions

Scheme A:

The total amount in both the years together = Rs.4500

Since, the amount invested in 2nd year is 25% more than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = 100: 125 = 4: 5

The amount invested in 1st year = $4500 \times (4/9) = \text{Rs.}2000$

And the amount invested in 2nd year = $4500 - 2000 = \text{Rs.}2500$

Let the rate of interest for 1st year = a%

So,

$$(2000 \times a)/100 = 170$$

$$a = 8.5\%$$

Since, the rate of interest for 1st year = 8.5%

So, the rate of interest for 2nd year = $8.5 + 3.5 = 12\%$

Scheme B:

The total amount in both the years together = Rs.4000

Since, the amount invested in 2nd year is 50% more than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = 100: 150 = 2: 3

The amount invested in 1st year = $4000 \times (2/5) = \text{Rs.}1600$

And the amount invested in 2nd year = $4000 - 1600 = \text{Rs.}2400$

@QuizByMickey

Let the rate of interest for 1st year = b%

So,

$$(1600 \times b)/100 = 240$$

$$b = 15\%$$

Since, the rate of interest for 1st year = 15%

So, the rate of interest for 2nd year = $15 - 10 = 5\%$

Scheme C:

The total amount in both the years together = Rs.3500

Since, the amount invested in 2nd year is 25% less than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = 100: 75 = 4: 3

The amount invested in 1st year = $3500 \times (4/7) = \text{Rs.}2000$

And the amount invested in 2nd year = $3500 - 2000 = \text{Rs.}1500$

Let the rate of interest for 1st year = c%

So,

$$(2000 \times c)/100 = 150$$

$$c = 7.5\%$$

Since, the rate of interest for 1st year = 7.5%

So, the rate of interest for 2nd year = $7.5 + 2.5 = 10\%$

Scheme D:

The total amount in both the years together = Rs.6000

Since, the amount invested in 2nd year is 40% more than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = 100: 140 = 5: 7

The amount invested in 1st year = $6000 \times (5/12) = \text{Rs.}2500$

And the amount invested in 2nd year = $6000 - 2500 = \text{Rs.}3500$

Let the rate of interest for 1st year = d%

So,

$$(2500 \times d)/100 = 150$$

$$d = 6\%$$

Since, the rate of interest for 1st year = 6%

So, the rate of interest for 2nd year = $6 + 6 = 12\%$

Scheme E:

The total amount in both the years together = Rs.3500

Since, the amount invested in 2nd year is 60% loss than that invested in 1st year.

@QuizByMickey

So, the ratio of the amount invested in 1st year to that invested in 2nd year = 100: 40 = 5: 2

The amount invested in 1st year = $3500 * (5/7) = \text{Rs.}2500$

And the amount invested in 2nd year = $3500 - 2500 = \text{Rs.}1000$

Let the rate of interest for 1st year = $e\%$

So,

$$(2500 * e)/100 = 200$$

$$e = 8\%$$

Since, the rate of interest for 1st year = 8%

So, the rate of interest for 2nd year = $8 + 6 = 14\%$

So, the table

Scheme	An amount invested in 1 st year (in ₹)	An amount invested in 2 nd year (in ₹)	The rate of interest for 1 st year	The rate of interest for 2 nd year
A	2000	2500	8.5%	12%
B	1600	2400	15%	5%
C	2000	1500	7.5%	10%
D	2500	3500	6%	12%
E	2500	1000	8%	14%

Answer: C

Since, the total amount invested in all the five years together in 1st year = $2000 + 1600 + 2000 + 2500 + 2500 = \text{Rs.}10600$

And the total amount invested in all the five years together in 2nd year = $2500 + 2400 + 1500 + 3500 + 1000 = \text{Rs.}10900$

So, the total SI received by scheme E after 2 years would have been:

$$= [(10600 * 8)/100] + [(10600 + 10900) * 14]/100]$$

$$= 848 + 3010$$

$$= \text{Rs.}3858$$

9. Questions

Scheme A:

The total amount in both the years together = Rs.4500

@QuizByMickey

Since, the amount invested in 2nd year is 25% more than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = 100: 125 = 4: 5

The amount invested in 1st year = $4500 \times (4/9) = \text{Rs.}2000$

And the amount invested in 2nd year = $4500 - 2000 = \text{Rs.}2500$

Let the rate of interest for 1st year = a%

So,

$$(2000 \times a)/100 = 170$$

$$a = 8.5\%$$

Since, the rate of interest for 1st year = 8.5%

So, the rate of interest for 2nd year = $8.5 + 3.5 = 12\%$

Scheme B:

The total amount in both the years together = Rs.4000

Since, the amount invested in 2nd year is 50% more than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = 100: 150 = 2: 3

The amount invested in 1st year = $4000 \times (2/5) = \text{Rs.}1600$

And the amount invested in 2nd year = $4000 - 1600 = \text{Rs.}2400$

Let the rate of interest for 1st year = b%

So,

$$(1600 \times b)/100 = 240$$

$$b = 15\%$$

Since, the rate of interest for 1st year = 15%

So, the rate of interest for 2nd year = $15 - 10 = 5\%$

Scheme C:

The total amount in both the years together = Rs.3500

Since, the amount invested in 2nd year is 25% less than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = 100: 75 = 4: 3

The amount invested in 1st year = $3500 \times (4/7) = \text{Rs.}2000$

@QuizByMickey

And the amount invested in 2nd year = $3500 - 2000 = \text{Rs.}1500$

Let the rate of interest for 1st year = $c\%$

So,

$$(2000 * c)/100 = 150$$

$$c = 7.5\%$$

Since, the rate of interest for 1st year = 7.5%

So, the rate of interest for 2nd year = $7.5 + 2.5 = 10\%$

Scheme D:

The total amount in both the years together = $\text{Rs.}6000$

Since, the amount invested in 2nd year is 40% more than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = $100: 140 = 5: 7$

The amount invested in 1st year = $6000 * (5/12) = \text{Rs.}2500$

And the amount invested in 2nd year = $6000 - 2500 = \text{Rs.}3500$

Let the rate of interest for 1st year = $d\%$

So,

$$(2500 * d)/100 = 150$$

$$d = 6\%$$

Since, the rate of interest for 1st year = 6%

So, the rate of interest for 2nd year = $6 + 6 = 12\%$

Scheme E:

The total amount in both the years together = $\text{Rs.}3500$

Since, the amount invested in 2nd year is 60% loss than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = $100: 40 = 5: 2$

The amount invested in 1st year = $3500 * (5/7) = \text{Rs.}2500$

And the amount invested in 2nd year = $3500 - 2500 = \text{Rs.}1000$

Let the rate of interest for 1st year = $e\%$

So,

$$(2500 * e)/100 = 200$$

$$e = 8\%$$

Since, the rate of interest for 1st year = 8%

So, the rate of interest for 2nd year = 8 + 6 = 14%

So, the table

Scheme	An amount invested in 1 st year (in ₹)	An amount invested in 2 nd year (in ₹)	The rate of interest for 1 st year	The rate of interest for 2 nd year
A	2000	2500	8.5%	12%
B	1600	2400	15%	5%
C	2000	1500	7.5%	10%
D	2500	3500	6%	12%
E	2500	1000	8%	14%

Answer: B

Since, the sum of the interest rates of all the five schemes for 1st year = 8.5 + 15 + 7.5 + 6 + 8 = 45

So, the degree distribution of the interest rate of scheme A for 1st year = 360 * (8.5/45) = 68°

And the degree distribution of the interest rate of scheme B for 1st year = 360 * (15/45) = 120°

Since, the value of N = 120 – 68 = 52

So, the total degree distribution of the interest rates of schemes C, D and E together for 1st year = 360 * [(7.5 + 6 + 8)/45] = 172° = (3N + 16)°

10. Questions

Scheme A:

The total amount in both the years together = Rs.4500

Since, the amount invested in 2nd year is 25% more than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = 100: 125 = 4: 5

The amount invested in 1st year = 4500 * (4/9) = Rs.2000

And the amount invested in 2nd year = 4500 – 2000 = Rs.2500

Let the rate of interest for 1st year = a%

So,

@QuizByMickey

$$(2000 * a)/100 = 170$$

$$a = 8.5\%$$

Since, the rate of interest for 1st year = 8.5%

So, the rate of interest for 2nd year = 8.5 + 3.5 = 12%

Scheme B:

The total amount in both the years together = Rs.4000

Since, the amount invested in 2nd year is 50% more than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = 100: 150 = 2: 3

The amount invested in 1st year = 4000 * (5/2) = Rs.1600

And the amount invested in 2nd year = 4000 – 1600 = Rs.2400

Let the rate of interest for 1st year = b%

So,

$$(1600 * b)/100 = 240$$

$$b = 15\%$$

Since, the rate of interest for 1st year = 15%

So, the rate of interest for 2nd year = 15 - 10 = 5%

Scheme C:

The total amount in both the years together = Rs.3500

Since, the amount invested in 2nd year is 25% less than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = 100: 75 = 4: 3

The amount invested in 1st year = 3500 * (4/7) = Rs.2000

And the amount invested in 2nd year = 3500 – 2000 = Rs.1500

Let the rate of interest for 1st year = c%

So,

$$(2000 * c)/100 = 150$$

$$c = 7.5\%$$

Since, the rate of interest for 1st year = 7.5%

@QuizByMickey

So, the rate of interest for 2nd year = $7.5 + 2.5 = 10\%$

Scheme D:

The total amount in both the years together = Rs.6000

Since, the amount invested in 2nd year is 40% more than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = $100: 140 = 5: 7$

The amount invested in 1st year = $6000 * (5/12) = \text{Rs.}2500$

And the amount invested in 2nd year = $6000 - 2500 = \text{Rs.}3500$

Let the rate of interest for 1st year = $d\%$

So,

$$(2500 * d)/100 = 150$$

$$d = 6\%$$

Since, the rate of interest for 1st year = 6%

So, the rate of interest for 2nd year = $6 + 6 = 12\%$

Scheme E:

The total amount in both the years together = Rs.3500

Since, the amount invested in 2nd year is 60% loss than that invested in 1st year.

So, the ratio of the amount invested in 1st year to that invested in 2nd year = $100: 40 = 5: 2$

The amount invested in 1st year = $3500 * (5/7) = \text{Rs.}2500$

And the amount invested in 2nd year = $3500 - 2500 = \text{Rs.}1000$

Let the rate of interest for 1st year = $e\%$

So,

$$(2500 * e)/100 = 200$$

$$e = 8\%$$

Since, the rate of interest for 1st year = 8%

So, the rate of interest for 2nd year = $8 + 6 = 14\%$

So, the table

@QuizByMickey

Scheme	An amount invested in 1 st year (in ₹)	An amount invested in 2 nd year (in ₹)	The rate of interest for 1 st year	The rate of interest for 2 nd year
A	2000	2500	8.5%	12%
B	1600	2400	15%	5%
C	2000	1500	7.5%	10%
D	2500	3500	6%	12%
E	2500	1000	8%	14%

Answer: C

From the above question:

The value of N = 52

From X:

The SI received from scheme B on 2nd year = $(4000 * 5)/100 = \text{Rs.}200$

Since, the value of $[4(N - 2)] = 4(52 - 2) = 200$

So, X is true.

From Y:

Since, the SI received from scheme C after 1 year = $(2000 * 7.5)/100 = \text{Rs.}150$

And the SI received from scheme C after 2 years = $(3500 * 10)/100 = \text{Rs.}350$

So, the required difference = $350 - 150 = \text{Rs.}200$

Since, the value of $(3N + 44) = (3 * 52 + 44) = 200$

So, Y is true.

From Z:

The sum of the interest rates of all the five schemes for 2nd year = $12 + 5 + 10 + 12 + 14 = 53$

Since, the value of $(N + 1) = 52 + 1 = 53$

So, Z is true.

Hence, all are true.

11. Questions

Answer: D

The distance between bikes A and C = 160 m

The distance between bikes B and C = 120 m

Since, bike B is following bike C, which is following bike A.

@QuizByMickey

So, the distance between bikes A and B = $160 + 120 = 280$

Let the speed of bike B = 'x' m/s

Since, the speed of bike A = $64.8 \text{ km/h} = 64.8 * (5/18) = 18 \text{ m/s}$

And bike B crosses bike A in 40 seconds.

So,

$$(x - 18) = 280/40$$

$$x = 25$$

The speed of bike B = 25 m/s

From P:

Since, the speed of bike B = 25 m/s

So, the time taken by bike B to cross a 450 m long bridge = $(450/25) = 18$ seconds

So, P can be determined.

From Q:

Since, we don't know the speed of bike C.

So, Q can't be determined.

From R:

Since, we don't know the speed of bike C.

So, R can't be determined.

Hence, only Q and R can be determined

12. Questions

Answer: E

Let the CP of the article = Rs.10x

So, the MP of the article = 160% of $10x$ = Rs.16x

And the SP of the article = Rs.(10x + 25)

If the shopkeeper sold one unit of the article at 40% profit on its cost price and one unit at 20% loss on its cost price:

The total SP of both the units of the article together:

$$140\% \text{ of } 10x + 80\% \text{ of } 10x$$

$$\text{Rs.}22x$$

Since, the total profit = Rs.25

So,

@QuizByMickey

$$22x - (2 * 10x) = 25$$

$$x = 12.5$$

Since, the MP of the article = $16 * 12.5 = \text{Rs.}200$

And the SP of the article = $(10 * 12.5 + 25) = \text{Rs.}150$

So, the required percentage = $[(200 - 150)/200] * 100 = 25\%$

13. Questions

Answer: A

Let the quantity of water in the mixture = 'x' L

So, the quantity of milk in the mixture = $(x + 40)$ L

Since, the SP of the mixture = Rs.81 per L

So, the CP of the mixture = $81 * (100/125) = \text{Rs.}64.8$ per L

Since, the cost of pure milk = Rs.72 per L

So,

$$[(x + 40) * 72]/(x + 40 + x) = 64.8$$

$$72x + 2880 = 129.6x + 2592$$

$$57.6x = 288$$

$$x = 5$$

The quantity of milk in the mixture = $5 + 40 = 45$ L

The quantity of water in the mixture = 5 L

The cost of pure milk = Rs.72 per L

From P:

When 10 L water is added in the mixture:

The cost of the mixture = $(45 * 72)/(45 + 15) = \text{Rs.}54$ per L

So, the values in P can be filled in the blanks.

From Q:

When 4 L water is added in the mixture:

The cost of the mixture = $(45 * 72)/(45 + 9) = \text{Rs.}60$ per L

So, the values in Q can be filled in the blanks.

From R:

When 14 L water is added in the mixture:

The cost of the mixture = $(45 * 72)/(45 + 19) = \text{Rs.}50.625$ per L

@QuizByMickey

So, values in R can't be filled in the blanks.

Hence, only values of R can't be filled in the blanks.

14. Questions

Answer: B

Since, the average age of the family = 24 years

$$\text{So, } (A + B + C)/3 = 24$$

$$A + B + C = 72 \text{ -----(1)}$$

Since, the difference between A's and B's present ages is equal to C's present age and A is elder to B.

So,

$$A - B = C \text{ -----(2)}$$

From equations (1) and (2):

$$A + B + A - B = 72$$

$$A = 36$$

From equation (1):

$$36 + B + C = 72$$

$$B + C = 36$$

From I:

If B's present age is 30 years:

$$\text{Then C's present age} = 36 - 30 = 6 \text{ years}$$

$$\text{And C's age after 6 years} = P = 6 + 6 = 12 \text{ years}$$

From II:

If B's present age is 20 years:

$$\text{Then C's present age} = 36 - 20 = 16 \text{ years}$$

$$\text{And C's age 8 years ago} = Q = 16 - 8 = 8 \text{ years}$$

From III:

If B's present age is 24 years:

$$\text{Then C's present age} = R = 36 - 24 = 12 \text{ years}$$

$$\text{Required ratio} = P: Q: R = 12: 8: 12 = 3: 2: 3$$

15. Questions

Answer: B

The time, in which A alone can complete the work = 18 hours

@QuizByMickey

The time, in which B alone can complete the work = 45 hours

Let the total work = LCM of 18 and 45 = 90 units

So, the work completed by A alone in 1 hour = $90/18 = 5$ units

And the work completed by B alone in 1 hour = $90/45 = 2$ units

Since, the part of work completed by B alone in 10 hours = $2 * 10 = 20$ units

So, the part of work completed by A and B together in 't' hours = $90 - 20 = 70$ units

Since, the part of work completed by A and B together in 1 hour = $5 + 2 = 7$ units

So, the value of 't' = $70/7 = 10$ hours

And the time, in which C alone can complete the work = $3 * 10 = 30$ hours

Since, the part of work completed by A, B and C together in 1 hour:

$$= (1/18) + (1/45) + (1/30)$$

$$= (5 + 2 + 3)/90$$

$$= 1/9$$

So, the time, in which A, B and C together can complete the work = 9 hours = (t - 1) hours

16. Questions

Since 'x' (< 100) is an even multiple of 11.

Possible values of 'x' = 22, 44, 66, and 86.

Out of which only 66 is 2 more than a perfect square of an integer.

So, x = 66

Total output items = 132

In phase 4, $1/5^{\text{th}}$ are discarded, which means total units reached at phase 4 of the testing = $132 * 5/(5 - 1) = 132 * 5/4 = 165$

Now, 165 units are reached at phase 4 from phase 2 (Round 2).

Which means total units reached at phase 2 (Round 2) of the testing = $165 * 100/(100 - 25) = 165 * 100/75 = 220$

Now, 220 units are reached at phase 2 (Round 2) from phase 3.

Which means total units reached at phase 3 of the testing = $220 * 15/(15 - p) = 3300/(15 - p)$

Now, ' $3300/(15 - p)$ ' units are reached at phase 3 from phase 2 (Round 1).

Which means total units reached at phase 2 (Round 1) of the testing = $3300/(15 - p) * 100/(100 - 25) = 4400/(15 - p)$

Now, ' $4400/(15 - p)$ ' units are reached at phase 2 (Round 1) from phase 1.

Which means total manufactured units = $[4400/(15 - p)] + 66 + 3(p - 1) = [4400/(15 - p)] + 63 + 3p$

Difference between total units of items A and B/Total manufactured unit of items A and B together = $(3 - 2)/(3 + 2)$

$$(23p + 3)/[4400/(15 - p) + 63 + 3p] = 1/5$$

$$[(23p + 3)(15 - p)]/[(4400 + (63 + 3p)(15 - p))] = 1/5$$

$$5(23p + 3)(15 - p) = 4400 + (63 + 3p)(15 - p)$$

$$-115p^2 + 1710p + 225 = 4400 - 3p^2 - 18p + 945$$

$$112p^2 - 1728p + 5120 = 0$$

$$7p^2 - 108p + 320 = 0$$

$$7p^2 - 28p - 80p + 320 = 0$$

$$7p(p - 4) - 80(p - 4) = 0$$

$$(p - 4)(7p - 80) = 0$$

$p = 4$ and $80/7$ [Since units cannot be fraction,]

So,

$$p = 4$$

Total units of manufactured types A and B together = $4400/(15 - p) + 63 + 3p = 4400/11 + 63 + 12 = 400 + 75 = 475$

Total units of manufactured type A = $475 * 3/5 = 285$

Total units of manufactured type B = $475 * 2/5 = 190$

Units discarded in phase 1 = $x + 3(p - 1) = 66 + 9 = 75$

Units discarded in phase 2 (Round 1) = 25% of $(475 - 75) = 25\%$ of $400 = 100$

Units discarded in phase 3 = $p/15 * (400 - 100) = 4/15 * 300 = 80$

Units discarded in phase 2 (Round 2) = 25% of $(300 - 80) = 25\%$ of $220 = 55$

Units discarded in phase 4 = $1/5 * (220 - 55) = 1/5 * 165 = 33$

Table given below shows the discarded and passed items in each testing phases:

Phases	Number of units reached for testing	Total units passed the testing	Total units discarded
Phase 1	475	400	75
Phase 2 (Round 1)	400	300	100
Phase 3	300	220	80
Phase 2 (Round 2)	220	165	55
Phase 4	165	132	33

Answer: C

According to the question:

Total discarded units of both type of items together from phase 1 of the testing to phase 2 (Round 2) of the testing = $75 + 100 + 80 + 55 = 310$

17. Questions

Since 'x' (< 100) is an even multiple of 11.

Possible values of 'x' = 22, 44, 66, and 86.

Out of which only 66 is 2 more than a perfect square of an integer.

So, $x = 66$

Total output items = 132

In phase 4, $1/5^{\text{th}}$ are discarded, which means total units reached at phase 4 of the testing = $132 * 5/(5 - 1) = 132 * 5/4 = 165$

Now, 165 units are reached at phase 4 from phase 2 (Round 2).

Which means total units reached at phase 2 (Round 2) of the testing = $165 * 100/(100 - 25) = 165 * 100/75 = 220$

Now, 220 units are reached at phase 2 (Round 2) from phase 3.

Which means total units reached at phase 3 of the testing = $220 * 15/(15 - p) = 3300/(15 - p)$

Now, ' $3300/(15 - p)$ ' units are reached at phase 3 from phase 2 (Round 1).

Which means total units reached at phase 2 (Round 1) of the testing = $3300/(15 - p) * 100/(100 - 25) = 4400/(15 - p)$

Now, ' $4400/(15 - p)$ ' units are reached at phase 2 (Round 1) from phase 1.

Which means total manufactured units = $[4400/(15 - p)] + 66 + 3(p - 1) = [4400/(15 - p)] + 63 + 3p$

Difference between total units of items A and B/Total manufactured unit of items A and B together = $(3 - 2)/(3 + 2)$

$$(23p + 3)/[4400/(15 - p) + 63 + 3p] = 1/5$$

$$[(23p + 3)(15 - p)]/[(4400 + (63 + 3p)(15 - p))] = 1/5$$

$$5(23p + 3)(15 - p) = 4400 + (63 + 3p)(15 - p)$$

$$-115p^2 + 1710p + 225 = 4400 - 3p^2 - 18p + 945$$

$$112p^2 - 1728p + 5120 = 0$$

$$7p^2 - 108p + 320 = 0$$

$$7p^2 - 28p - 80p + 320 = 0$$

@QuizByMickey

$$7p(p - 4) - 80(p - 4) = 0$$

$$(p - 4)(7p - 80) = 0$$

$$p = 4 \text{ and } 80/7 \text{ [Since units cannot be fraction.]}$$

So,

$$p = 4$$

$$\text{Total units of manufactured types A and B together} = 4400/(15 - p) + 63 + 3p = 4400/11 + 63 + 12 = 400 + 75 = 475$$

$$\text{Total units of manufactured type A} = 475 * 3/5 = 285$$

$$\text{Total units of manufactured type B} = 475 * 2/5 = 190$$

$$\text{Units discarded in phase 1} = x + 3(p - 1) = 66 + 9 = 75$$

$$\text{Units discarded in phase 2 (Round 1)} = 25\% \text{ of } (475 - 75) = 25\% \text{ of } 400 = 100$$

$$\text{Units discarded in phase 3} = p/15 * (400 - 100) = 4/15 * 300 = 80$$

$$\text{Units discarded in phase 2 (Round 2)} = 25\% \text{ of } (300 - 80) = 25\% \text{ of } 220 = 55$$

$$\text{Units discarded in phase 4} = 1/5 * (220 - 55) = 1/5 * 165 = 33$$

Table given below shows the discarded and passed items in each testing phases:

Phases	Number of units reached for testing	Total units passed the testing	Total units discarded
Phase 1	475	400	75
Phase 2 (Round 1)	400	300	100
Phase 3	300	220	80
Phase 2 (Round 2)	220	165	55
Phase 4	165	132	33

Answer: E

The total units passed in phase 2 (Round 1) of the testing = 300

The total units of type A passed in phase 2 (Round 1) of the testing = 66% of 300 = 198

The total units of type B passed in phase 2 (Round 1) of the testing = 300 - 198 = 102

Required difference = 198 - 102 = 96

18. Questions

Since 'x' (< 100) is an even multiple of 11.

Possible values of 'x' = 22, 44, 66, and 86.

Out of which only 66 is 2 more than a perfect square of an integer.

So, x = 66

Total output items = 132

In phase 4, $\frac{1}{5}$ th are discarded, which means total units reached at phase 4 of the testing = $132 * \frac{5}{(5-1)} = 132 * \frac{5}{4} = 165$

Now, 165 units are reached at phase 4 from phase 2 (Round 2).

Which means total units reached at phase 2 (Round 2) of the testing = $165 * \frac{100}{(100-25)} = 165 * \frac{100}{75} = 220$

Now, 220 units are reached at phase 2 (Round 2) from phase 3.

Which means total units reached at phase 3 of the testing = $220 * \frac{15}{(15-p)} = \frac{3300}{(15-p)}$

Now, ' $\frac{3300}{(15-p)}$ ' units are reached at phase 3 from phase 2 (Round 1).

Which means total units reached at phase 2 (Round 1) of the testing = $\frac{3300}{(15-p)} * \frac{100}{(100-25)} = \frac{4400}{(15-p)}$

Now, ' $\frac{4400}{(15-p)}$ ' units are reached at phase 2 (Round 1) from phase 1.

Which means total manufactured units = $[\frac{4400}{(15-p)}] + 66 + 3(p-1) = [\frac{4400}{(15-p)}] + 63 + 3p$

Difference between total units of items A and B/Total manufactured unit of items A and B together = $(3-2)/(3+2)$

$$(23p+3)/[\frac{4400}{(15-p)} + 63 + 3p] = \frac{1}{5}$$

$$[(23p+3)(15-p)]/[(4400 + (63 + 3p)(15-p))] = \frac{1}{5}$$

$$5(23p+3)(15-p) = 4400 + (63 + 3p)(15-p)$$

$$-115p^2 + 1710p + 225 = 4400 - 3p^2 - 18p + 945$$

$$112p^2 - 1728p + 5120 = 0$$

$$7p^2 - 108p + 320 = 0$$

$$7p^2 - 28p - 80p + 320 = 0$$

$$7p(p-4) - 80(p-4) = 0$$

$$(p-4)(7p-80) = 0$$

$p = 4$ and $80/7$ [Since units cannot be fraction.]

So,

$$p = 4$$

Total units of manufactured types A and B together = $\frac{4400}{(15-p)} + 63 + 3p = \frac{4400}{11} + 63 + 12 = 400 + 75 = 475$

Total units of manufactured type A = $475 * \frac{3}{5} = 285$

Total units of manufactured type B = $475 * \frac{2}{5} = 190$

@QuizByMickey

Units discarded in phase 1 = $x + 3(p - 1) = 66 + 9 = 75$

Units discarded in phase 2 (Round 1) = 25% of $(475 - 75) = 25\%$ of 400 = 100

Units discarded in phase 3 = $p/15 * (400 - 100) = 4/15 * 300 = 80$

Units discarded in phase 2 (Round 2) = 25% of $(300 - 80) = 25\%$ of 220 = 55

Units discarded in phase 4 = $1/5 * (220 - 55) = 1/5 * 165 = 33$

Table given below shows the discarded and passed items in each testing phases:

Phases	Number of units reached for testing	Total units passed the testing	Total units discarded
Phase 1	475	400	75
Phase 2 (Round 1)	400	300	100
Phase 3	300	220	80
Phase 2 (Round 2)	220	165	55
Phase 4	165	132	33

Answer: B

'y' is 2-digit (> 20) prime number '1' more than a perfect square)

2-digit (> 20) prime numbers = 25, 36, 49, 64, and 81.

Out of which only 36 is feasible because 37 is a prime number.

SO, $y = 37$

Total items discarded in phase 5 of the testing = $y = 37$

Total items passed after last phase of the testing = $132 - 37 = 95$

Total manufactured items = 475

Required percent = $95/475 * 100$

= 20%

19. Questions

Since 'x' (< 100) is an even multiple of 11.

Possible values of 'x' = 22, 44, 66, and 86.

Out of which only 66 is 2 more than a perfect square of an integer.

So, $x = 66$

Total output items = 132

In phase 4, $1/5^{\text{th}}$ are discarded, which means total units reached at phase 4 of the testing = $132 * 5/(5 - 1) = 132 * 5/4 = 165$

@QuizByMickey

Now, 165 units are reached at phase 4 from phase 2 (Round 2).

Which means total units reached at phase 2 (Round 2) of the testing = $165 * 100 / (100 - 25) = 165 * 100 / 75 = 220$

Now, 220 units are reached at phase 2 (Round 2) from phase 3.

Which means total units reached at phase 3 of the testing = $220 * 15 / (15 - p) = 3300 / (15 - p)$

Now, ' $3300 / (15 - p)$ ' units are reached at phase 3 from phase 2 (Round 1).

Which means total units reached at phase 2 (Round 1) of the testing = $3300 / (15 - p) * 100 / (100 - 25) = 4400 / (15 - p)$

Now, ' $4400 / (15 - p)$ ' units are reached at phase 2 (Round 1) from phase 1.

Which means total manufactured units = $[4400 / (15 - p)] + 66 + 3(p - 1) = [4400 / (15 - p)] + 63 + 3p$

Difference between total units of items A and B / Total manufactured unit of items A and B together = $(3 - 2) / (3 + 2)$

$$(23p + 3) / [4400 / (15 - p) + 63 + 3p] = 1/5$$

$$[(23p + 3)(15 - p)] / [(4400 + (63 + 3p)(15 - p))] = 1/5$$

$$5(23p + 3)(15 - p) = 4400 + (63 + 3p)(15 - p)$$

$$-115p^2 + 1710p + 225 = 4400 - 3p^2 - 18p + 945$$

$$112p^2 - 1728p + 5120 = 0$$

$$7p^2 - 108p + 320 = 0$$

$$7p^2 - 28p - 80p + 320 = 0$$

$$7p(p - 4) - 80(p - 4) = 0$$

$$(p - 4)(7p - 80) = 0$$

$p = 4$ and $80/7$ [Since units cannot be fraction.]

So,

$$p = 4$$

Total units of manufactured types A and B together = $4400 / (15 - p) + 63 + 3p = 4400 / 11 + 63 + 12 = 400 + 75 = 475$

Total units of manufactured type A = $475 * 3/5 = 285$

Total units of manufactured type B = $475 * 2/5 = 190$

Units discarded in phase 1 = $x + 3(p - 1) = 66 + 9 = 75$

Units discarded in phase 2 (Round 1) = 25% of $(475 - 75) = 25\% \text{ of } 400 = 100$

Units discarded in phase 3 = $p/15 * (400 - 100) = 4/15 * 300 = 80$

@QuizByMickey

Units discarded in phase 2 (Round 2) = 25% of (300 – 80) = 25% of 220 = 55

Units discarded in phase 4 = $\frac{1}{5} * (220 - 55) = \frac{1}{5} * 165 = 33$

Table given below shows the discarded and passed items in each testing phases:

Phases	Number of units reached for testing	Total units passed the testing	Total units discarded
Phase 1	475	400	75
Phase 2 (Round 1)	400	300	100
Phase 3	300	220	80
Phase 2 (Round 2)	220	165	55
Phase 4	165	132	33

Answer: E

Total units of manufactured item A = $475 * \frac{3}{5} = 285$

Total units of manufactured item B = $475 * \frac{2}{5} = 190$

Total passed units of type A in phase 1 = $400 * \frac{11}{20} = 220$

Total passed units of type B in phase 1 = $400 - 220 = 180$

Required ratio = $(285 - 220) : (190 - 180)$

= 65: 10

= 13: 2

20. Questions

Since 'x' (< 100) is an even multiple of 11.

Possible values of 'x' = 22, 44, 66, and 86.

Out of which only 66 is 2 more than a perfect square of an integer.

So, x = 66

Total output items = 132

In phase 4, $\frac{1}{5}^{\text{th}}$ are discarded, which means total units reached at phase 4 of the testing = $132 * \frac{5}{(5 - 1)} = 132 * \frac{5}{4} = 165$

Now, 165 units are reached at phase 4 from phase 2 (Round 2).

Which means total units reached at phase 2 (Round 2) of the testing = $165 * \frac{100}{(100 - 25)} = 165 * \frac{100}{75} = 220$

Now, 220 units are reached at phase 2 (Round 2) from phase 3.

Which means total units reached at phase 3 of the testing = $220 * \frac{15}{(15 - p)} = \frac{3300}{(15 - p)}$

Now, ' $\frac{3300}{(15 - p)}$ ' units are reached at phase 3 from phase 2 (Round 1).

@QuizByMickey

Which means total units reached at phase 2 (Round 1) of the testing = $3300/(15 - p) * 100/(100 - 25) = 4400/(15 - p)$

Now, '4400/(15 - p)' units are reached at phase 2 (Round 1) from phase 1.

Which means total manufactured units = $[4400/(15 - p)] + 66 + 3(p - 1) = [4400/(15 - p)] + 63 + 3p$

Difference between total units of items A and B/Total manufactured unit of items A and B together = $(3 - 2)/(3 + 2)$

$$(23p + 3)/[4400/(15 - p) + 63 + 3p] = 1/5$$

$$[(23p + 3)(15 - p)]/[(4400 + (63 + 3p)(15 - p))] = 1/5$$

$$5(23p + 3)(15 - p) = 4400 + (63 + 3p)(15 - p)$$

$$-115p^2 + 1710p + 225 = 4400 - 3p^2 - 18p + 945$$

$$112p^2 - 1728p + 5120 = 0$$

$$7p^2 - 108p + 320 = 0$$

$$7p^2 - 28p - 80p + 320 = 0$$

$$7p(p - 4) - 80(p - 4) = 0$$

$$(p - 4)(7p - 80) = 0$$

$p = 4$ and $80/7$ [Since units cannot be fraction.]

So,

$$p = 4$$

Total units of manufactured types A and B together = $4400/(15 - p) + 63 + 3p = 4400/11 + 63 + 12 = 400 + 75 = 475$

Total units of manufactured type A = $475 * 3/5 = 285$

Total units of manufactured type B = $475 * 2/5 = 190$

Units discarded in phase 1 = $x + 3(p - 1) = 66 + 9 = 75$

Units discarded in phase 2 (Round 1) = 25% of $(475 - 75) = 25\%$ of $400 = 100$

Units discarded in phase 3 = $p/15 * (400 - 100) = 4/15 * 300 = 80$

Units discarded in phase 2 (Round 2) = 25% of $(300 - 80) = 25\%$ of $220 = 55$

Units discarded in phase 4 = $1/5 * (220 - 55) = 1/5 * 165 = 33$

Table given below shows the discarded and passed items in each testing phases:

Phases	Number of units reached for testing	Total units passed the testing	Total units discarded
Phase 1	475	400	75
Phase 2 (Round 1)	400	300	100
Phase 3	300	220	80
Phase 2 (Round 2)	220	165	55
Phase 4	165	132	33

Answer: A

Total passed units of both the items in phase 4 = 132

Total passed units of both the items in phase 1 = 400

According to the question:

$$(10a + b)/100 = 132/400$$

$$(10a + b)/100 = 33/100$$

After comparing:

$$10a + b = 33$$

If $a = 1$

$$10 * 1 + b = 33$$

$$b = 23$$

It is not single digit number.

If $a = 2$

$$10 * 2 + b = 33$$

$$b = 13$$

It is not single digit number.

If $a = 3$

$$10 * 3 + b = 33$$

$$b = 3$$

It is single digit number.

So

$$a = b = 33$$

Now,

$$= (a + b)/ab$$

@QuizByMickey

$$= (3 + 3)/(3 * 3)$$

$$= 6/9$$

$$= 2/3$$

@QuizByMickey